2025

SPACE SETTLEMENT DESIGN COMPETITION

CENTRAL REGION
JOHNSON SPACE CENTER

InSimEd

Industry Simulation Education

Competitions Igniting Career Aspirations

February 7-9, 2025

Dear Students,

Welcome to the 2025 Central Space Settlement Design Competition. This competition has a rich history and we look forward to your success this year at the competition. It will be another step to help you determine your career path and what you may want to do for the next chapter of your life.

In the next three days you will be given the opportunity to create solutions for a complex situation. You will be engaged in real-world STEM (Science, Technology, Engineering, and Mathematics) experiences. How you work with peers, build partnerships, research and use resources will also determine your cooperative learning experience. We are excited to see your thinking and work put into practice over the next two days.

We would like to take a moment and thank all of our volunteers. This event would not be possible without their support. At some point this weekend, please take time to thank a professional that is volunteering! I would also like to thank the Gilruth Center for hosting our event.

We wish you the best this weekend and in the future. Challenge yourself, each other, and enjoy the process of learning. You are the future and we are looking forward to you creating a positive impact on society through this competition and in years to come. Make it a great weekend!

Sincerely,

Hayley Spears

Hayley Spears
Central Region Coordinator

And the rest of the InSimEd Team

2025 Central SSDC

Expectations of Behavior and Facility Rules

STUDENT CODE OF CONDUCT

InSimEd is dedicated to providing a professional, collegial, safe, and respectful environment for all participants, regardless of race, sex, religion or creed, national origin, disability, age, sexual orientation or other legally protected class as provided by law.

InSimEd will not tolerate harassment, discrimination, or any other unprofessional behavior. Inappropriate language and imagery are not acceptable in any of the Virtual Event spaces, social media, or any aspect of the Virtual Event.

Participants may not engage in any behavior that InSimEd deems to be potentially disruptive to the conduct of the competition, or to potentially infringe on the rights or safety of others, or otherwise violate the principles above. Any participants violating these rules may be sanctioned, expelled from the competition or subject to other consequences at the discretion of InSimEd.

Remember, you are a professional now. Your cooperation in following these instructions is expected and appreciated. Responsible behavior will help assure the future ability to continue the InSimEd competition.

THE GILRUTH CENTER CODE OF CONDUCT

See https://www.nasa.gov/Starport/WellnessAndFitness/FacilytyRulesandRegulations for more information.

The Gilruth Center facility is a multi-use recreation and conference facility in which a wide variety of services, programs, and events are conducted to enhance the morale and welfare of all JSC civil service and contractor employees, authorized visitors, and the local community.

The Gilruth Center is officially part of the NASA - Johnson Space Center. The Gilruth Center is located on federal property and is considered a federal facility. All rules, regulations and policies of a federal facility apply to the Gilruth Center facility, both indoor and outdoor.

In addition, the following Code of Conduct sets forth expectations for all participants and visitors who use the facility for any purpose.

Participants and visitors must adhere to the following:

- Act with courtesy and professionalism at all times.
- Comply with requests and direction from Starport staff, officials, instructors, and facilitators who are acting in the performance of their duties.
- Comply with any rules and guidance set forth for the particular program, event or class.

Participants and visitors must refrain from:

- Aggressive behavior in any form, including physical abuse, verbal abuse, threats, intimidation, harassment, coercion and/or conduct which threatens or endangers the health or safety of any person.
- Rude or argumentative behavior with staff, officials, and instructors or other participants or visitors.
- Disrupting or obstructing any program, event or class.
- Lewd, obscene or indecent conduct or expression, including profanity, or offensive remarks.
- Any action which constitutes an attempt to inflict, or the actual infliction of, or injury to other participants and/or staff.
- Willful damage or destruction to the facility or property.
- Forgery or sharing membership cards for access to the facility or access to programs, services or classes.
- Unauthorized entry to areas such as: staff offices or staff workspaces; gender opposite locker rooms; maintenance, equipment or storage rooms.
- Unauthorized use of facility computers or unauthorized adjusting of audio visual equipment.
- Photography and video taping of participants or visitors without prior approval of participants being photographed or taped and without prior approval from an authorized staff member.
- Unauthorized commercial activity – no person is allowed to post, advertise, instruct in private lessons or solicit individuals in the facility for personal services or for personal businesses that is not directly affiliated and approved through Starport.

All participants and visitors are required to report any violation of this Code of Conduct to a Starport staff member immediately. As needed, JSC Security may be called at any time to help enforce this Code of Conduct.

PRESS RELEASE

16 January 2068

FOUNDATION SOCIETY TO BUILD SETTLEMENT ON MARS

Foundation Society President Edwards Smith announced today that the organization's next space settlement, designated "Argonom", will be on the Martian surface. "Mars is quickly becoming a popular destination for research, goods manufacturing, and other economic activities", said Smith. "We will need facilities on the ground to support travel around the planet, for both people and for cargo."

Building the first settlement on Mars will come with challenges. Exploration teams report one particularly vexing Martian attribute. Mineralogist Lisa Cantu said "the dust is so fine it's called 'fines'. It's not as abrasive as lunar dust, but it's worse at getting into things. Lunar dust gets kicked up and falls right down again. Martian dust hangs in the atmosphere a while. When the wind blows--even at hundreds of kilometers per hour--it's no problem keeping control of the rovers or even standing up outside, because the atmosphere is so thin. But the wind picks up the dust and it gets into everything, at any height. It covers solar panels. It stuck in the creases and weaves of our suits. It blew into our airlocks with us, and followed us everywhere. It defied every attempt to get rid of it. It reminded me of Colorado River silt in the Grand Canyon--after a week it was irremovably in our clothes, in our beds, in our food, everywhere. And it's very effective at destroying machinery."

Human Factors engineer Jack Bayford said that despite the challenges of living on Mars, the people working in the research bases there want to stay. They like the reduced weight of 0.38 g, without the stability challenges of lunar 1/6 g, "and the scenery is incredible--features on Mars are ten times the size of features on Earth--mountains, canyons, everything". Just not lakes.

Smith said that Argonom is named after the mythological ship Argo, which protected its crew through a series of adventures to far-off places as they fulfilled their quest for the Golden Fleece. "Mars is as remote and dangerous to us as Colchis was to Jason, and we need all the help we can get to make this a prosperous voyage for the Foundation Society."

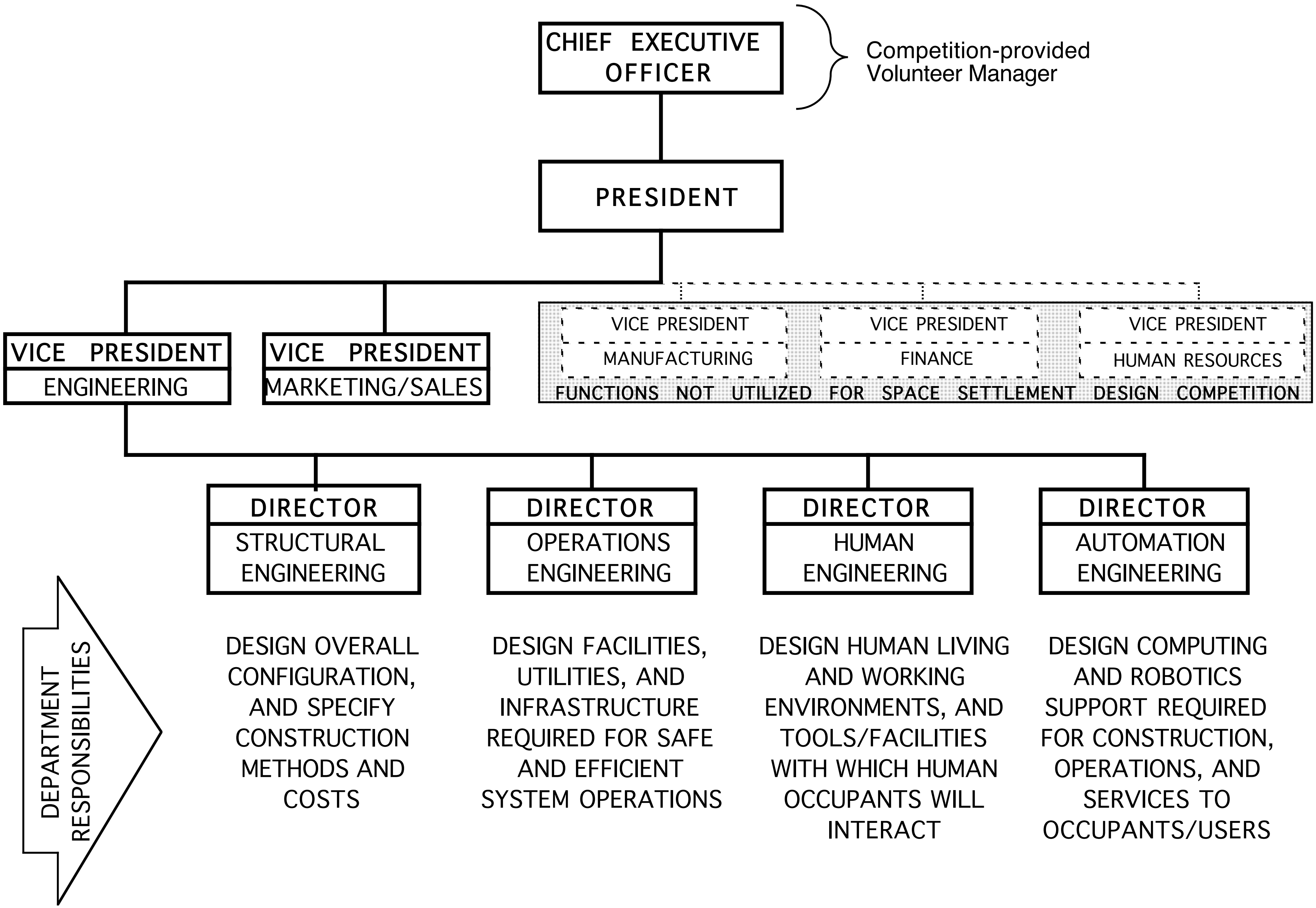
The Foundation Society's Chief Financial Officer Eileen Eduardo detailed the plans the Foundation has for business ventures on the Martian surface. "Argonom will serve as a hub for surface operations around the planet. Long-distance travelers arriving to Mars will stop at Aresam, then people and cargo will come down to Argonom before heading to their final destinations. As Argonom grows, some of the arrivals will become permanent residents." The Foundation is planning to have cargo storage and tracking capabilities as items arrive and leave the surface, and make their way around the planet. This includes sensitive items that need to stay at particular temperatures or atmospheric compositions, and of course, any containers opened will need to be protected from Martian fines.

"We will also offer transportation services, delivering cargo arriving on Mars to remote communities, or collecting items produced at other facilities, that will be stored at Argonom and shipped off-planet through Argonom's port facilities", said Eduardo. When this interplanetary transportation system is up and running, Argonom will also produce excess goods such as food or other commodities that the Foundation can manufacture, which will be sold to remote communities with no production capabilities.

Edwards Smith explained further, "It takes a long time to get from Earth to Mars, so there will likely be some amount of time between visits from Earth, as most travelers will take optimal routes to minimize fuel needs on the journey. Cargo storage capabilities with careful attention to detail and tracking of cargo will help keep all human activity on Mars running smoothly no matter when we expect a new batch of visitors."

Argonom will be another great engineering challenge for the Foundation Society, but with the right people dedicated to making it work, the project will help humanity colonize Mars.

COMPANY ORGANIZATION CHART



INTERACTIONS BETWEEN DEPARTMENTS

Each Engineering Department must provide and receive information to and from other Engineering Departments. Major products between Engineering Departments are:

PRODUCT	Supplied by	Provided to	TASK
Requirements for commercial, research, storage, transportation, and utilities facilities	Operations	Structural	Provide locations and space for operations facilities in design
Constraints on size and locations of operations facilities due to overall design requirements	Structural	Operations	Compromise to meet constraints while providing required facilities
Requirements for size and configuration of living areas	Human	Structural	Provide locations and space for living areas in design
Constraints on size and locations of living areas due to overall design considerations	Structural	Human	Compromise to meet constraints while providing required facilities
Space and location requirements for accommodation of computing facilities	Automation	Structural	Provide locations and space for computing facilities in design
Automation requirements during construction, and constraints on size and locations of computing areas due to overall design considerations	Structural	Automation	Design automated construction aids, and compromise to meet constraints while providing required services
Requirements for crew operation of settlement	Operations	Human	Design operations crew workspaces and environment, and user-friendly control systems
Requirements for support of human population living in an enclosed environment, and appropriate working conditions	Human	Operations	Develop requirements for utilities to meet resident needs, and adjust working conditions for human considerations
Computing and robotics requirements for operations acitivities	Operations	Automation	Design computing facilities and robotic equipment to meet operations needs.
Utilities requirements for computing and robotics equipment	Automation	Operations	Develop requirements for utilities to meet computing and robotics systems needs
Computing and robotics requirements for human activities	Human	Automation	Design computing facilities and robotics systems to meet resident and worker needs
Crew interaction requirements with computing and robotics equipment	Automation	Human	Design user workspaces, working conditions, and interfaces with computers and robots

DEPARTMENT DESCRIPTION: STRUCTURAL ENGINEERING

The Structural Engineering Department is responsible for defining the overall designs and configurations of projects built by the company. Engineers in this department are involved with the following areas of space settlement design:

Exteriors and basic configuration designs:

- Size
- Shape
- Accommodations for surrounding environment
  - Local gravity (or lack thereof; if rotating, interfaces with non-rotating volumes)
  - Atmosphere and weather (or lack thereof)
  - Surface conditions and topography (or lack thereof)
- Identify materials appropriate for each major part of the structure:
  - Consider effects of outside environment
    - Thermal factors, including expansion and contraction of facilities structures
    - Damage from debris or natural material impacts
    - Protection against radiation
  - Consider factors that influence required materials strength
  - Appropriateness of surface characteristics and opaqueness/transparency
  - Availability (to be identified by Operations Engineering)
- Atmosphere containment (where required)
- Lighting and provisions for day/night cycles
- Provisions for antennas, solar panels, radiators, and other infrastructure
- Provisions for facilities growth

Interior configuration:

- Allocation of volumes and acreage for different uses ("zoning")
- Design of suitable facilities for settlement activities (other depts define requirements), e.g.:
  - Residential
  - Commercial
  - Agricultural
  - Manufacturing, industrial
  - Port, commerce
- Distances and transportation routes between areas
- Provisions for installation of utilities (water, power, sewage, communications, ...)

Construction/assembly techniques:

- Assembly sequence, including on-site vs. pre-fab construction of components
- Construction functions to be performed by automated systems, and by people
- Size constraints imposed by existing cargo vehicle capacities
- Coordinate construction sequence with start-up sequence for beginning facility operations
- Determine overall facilities construction cost and schedule

System Engineering:

- Coordinate scheduling of all departments' inputs to design and construction process
- Consider proximity of related functions/activities, to enhance operational efficiency
- Safety accommodations:
  - Maintenance and repair
  - Redundant systems
  - Provisions for emergencies
  - Natural hazards analysis
  - Attempted sabotage analysis

DEPARTMENT DESCRIPTION: OPERATIONS ENGINEERING

The Operations Engineering Department is responsible for defining designs and requirements for infrastructure and utilities to support safe, cost-effective, and efficient operations of projects built by the company. Engineers in this department are involved with the following areas of space settlement design:

Location of facilities:
- Select orbit parameters and/or placement on planetary/lunar surface
- Provide rationale for location(s) selection(s), based on unique attributes of each site
- Design and/or define requirements for transportation service to/from the operating settlement:
  - Earth-to-orbit launch vehicles
  - Inter-orbital vehicles and nonterrestrial landers
  - Flight times and propellant requirements in space
  - Estimate frequency of visits by spacecraft, based on import/export/passenger expectations

Operations to support construction:
- Materials sources and transportation
- Equipment sources (and availability), and transportation
- Facilities infrastructure required during construction (e.g., accommodations for workers)
- Determine costs and schedule for building operations systems

Design utilities and infrastructure to support internal operations of the settlement:
- Electrical power:
  - Identify source(s) of power
  - Design power generation and distribution system(s)
- Transportation inside the facilities:
  - Design vehicles and define numbers of vehicles, for passengers and cargo
  - Define requirements for transportation corridors to accommodate vehicle traffic
- Communications system(s): design internal services and antenna(s) for external services
- Water acquisition and delivery systems
- Sewage and solid waste disposal systems
- Atmosphere maintenance and weather production
- Agriculture/food production: growing, harvesting, processing

Establish requirements to enhance overall facility design functionality and efficiency:
- Define facilities requirements to support different types of commercial activities, e.g.:
  - Extraterrestrial materials mining and refining
  - Manufacturing, for domestic and export use; capital and consumer goods
  - Extraterrestrial commerce center (e.g., port facilities)
  - Servicing/maintenance/repair/supplies base for visiting spacecraft
  - Storage requirements
    - Warehousing for commercial materials and products
    - Consumables
    - Spare parts and equipment
    - Fuel, including depot for space vehicles
  - Research
  - Tourism
  - Future expansion of commercial operations
- Define requirements for proximity of related functions (e.g., loading dock and warehousing)
- Establish personnel requirements for facilities operation and administration
  - Design work areas for facilities administrative staff
  - Define automated and other tools required for efficient facilities management
- Determine annual operating costs

DEPARTMENT DESCRIPTION: HUMAN ENGINEERING

The Human Engineering Department is responsible for determining requirements to insure the working effectiveness, safety, health, and comfort of the people who will be using facilities built by the company, and to complete designs of communities, workplaces, and homes. Engineers in this department are involved with the following areas of space settlement design:

Establish requirements for overall facility design to enhance livability for the residents:

- Define total volume of atmosphere and living space required for settlement population
- Define sizes of land areas required for activities to support human occupation
- Identify acceptable and unacceptable land uses near residential and community areas
- Define preferred accommodations for ambient gravity environment (or lack thereof)
  - Requirements for simulated gravity; specify acceleration environment(s)
  - Design facilities with attributes for other than 1 g
  - Handholds and tether systems in microgravity volumes
- Define preferred atmosphere conditions
  - Optimum pressure, mix of gases, and humidity
  - Weather/climate/seasons requirements
- Define preferred day/night and operating personnel work cycles

Define requirements for utilities and infrastructure to serve needs of residents:

- Water and sanitary facilities
- Communication and transportation services
- Food production to support community population
  - Quantity and variety of foods
  - Food distribution systems
- Non-food consumables to support community population
  - Types and quantities of consumer goods
  - Distribution systems
- Automated and other tools to enhance lifestyle

Design residential and community areas:

- Identify facilities and services that must be included in community plans, e.g.:
  - Schools
  - Medical facilities
  - Retail and professional services businesses
  - Food services
  - Entertainment and recreation
  - Facilities for meetings and civic/government functions
  - Accommodations for visitors
- Prepare map(s) of community(ies), showing locations and sizes of identified facilities
- Prepare map(s) of residential neighborhoods, indicating numbers and sizes of homes
- Provide examples of community design details to illustrate aesthetic appeal
  - Architectural styles and patterns
  - Attention to psychological factors
    - Impressions of openness and space; lines of sight
    - Greenbelts, natural surroundings
    - Create impressions that belie a high-density, enclosed environment
  - Designs/floor plans of homes
- Determine costs and schedule for constructing community and residential facilities

Develop designs and/or establish requirements to enhance human health and productivity in space:

- Protection against extraterrestrial environments (e.g., radiation exposure)
- Space suit design and operations
- Vehicles and tools
- Safety provisions in workplaces

DEPARTMENT DESCRIPTION: AUTOMATION ENGINEERING

The Automation Engineering Department is responsible for designing and providing computer and robotics services required to support operations of projects built by the company. Engineers in this department are involved with the following areas of space settlement design:

Identify requirements for computing support:

- Identify applications for which computers will be used, e.g.:
  - Settlement attitude control and station-keeping
  - Equipment and system monitoring and controlling, e.g.;
    - Environmental control and life support
    - Transportation system(s)
    - Manufacturing operations
    - Utilities management
  - Communications functions
  - Library/data base(s)
  - Support to businesses and commerce (e.g., sales, customer, and inventory records)
  - Support to product development and research
  - Household uses
- Determine computer size requirements based on anticipated use patterns
- Determine requirements for networking, internally and with other systems
- Determine requirements for "firewalls" to protect critical functions and data
- Identify locations where interfaces with computers will be required

Design computer system(s) to accommodate identified computing requirements:

- Numbers, types, and sizes of computers
- Computer networking configuration(s) (e.g., distributed network vs. central location)
- Operating systems and software; provisions for accommodating upgrades
- Redundancies and protections for critical functions and data
- Numbers and types of interfaces with computer(s) and computer system(s)
  - Human input/output functions
  - Display functions
  - Automated data collection and processing
  - Remote and hard-wired interfaces
- Commonality between computer system(s) and hardware components
- Facilities requirements to support computer operations (e.g., computer room)
- Estimate cost and schedule for developing, assembling, and installing computer system(s)
- Estimate annual maintenance requirements and costs

Identify requirements for robots:

- Identify applications for which robots will be used, e.g.:
  - Facilities construction
  - Maintenance and monitoring functions
  - Support to business, manufacturing, and commercial operations
  - Agriculture
  - Household uses
- Determine mobility and physical attributes required of robots for specific purposes

Design robot system(s) to accommodate identified requirements:

- Numbers, types, and sizes of robots
- Identify robot designs applicable for multiple applications
- Monitoring and commanding methods
- Illustrate robots, and show them in their application environments
- Facilities requirements to support robot operations (e.g., staging areas, pathways)
- Determine cost and schedule for developing, building, and installing robot system(s)
- Determine annual maintenance requirements and costs for robot system(s)

APPENDIX A: SUBCONTRACTORS

Major companies have access to a vast network of smaller companies making specialized products. They are eager to make or do anything they are capable of making or doing, for anyone who will pay. They frequently have contracts with competitors for the same proposal.

3D Logistics successfully 3D printed tools and components from a variety of feedstock materials at spaceport Freedom, and now has a production facility at Alexandriat. The company developed a 3D printing system that is flown on crewed spacecraft to enable replacing failed parts, eliminating most of the need for on-board logistics inventory.

BeamBuilders, Ltd. operates a manufacturing facility on a ferro-nickel asteroid in an Earthcrossing orbit, producing triangular trusses to customer-specified dimensions, at the rate of 500 linear feet per hour. Customers provide their own transportation of the structures, although limited assembly is permitted in the vicinity of this operation; orbital mechanics restricts practical delivery opportunities to twice per year. A standard triangular truss with 12-foot sections, suitable for zerog installations, typically sells for $1500 per linear foot. Custom orders are more expensive.

Blown Away specializes in making inflatable buildings per customer specifications, to enable quick construction of new communities on the Moon and in space. Although these structures are not intended for permanent or indefinite use, they provide shelter for residents to start new economic activities, until more durable solutions can be established. The company also has a product line of inflatable furniture, cheap to ship and simple to set up.

Bots4U offers purpose-built robots for home and office use. Currently available functions for robots are cleaning household surfaces, washing dishes, doing laundry, moving furniture, and fetching and stowing items for their owners. The company offers to build robots to customer specifications for space applications; robots built for use on Earth perform adequately in 0.38 g.

Bottom Cleaners recognized a critical need for long-term space habitation, first discussed in public at an NSS Space Settlement Summit in 2017: toilet paper. A system adequate for continuously supplying toilet paper for 1000 people fits in one CASSSC, and can be installed anywhere in a pressurized volume where power and water service are provided. The facility makes toilet paper out of plant fibers collected from agricultural and landscaping waste. Excess capacity can be used for making paper towels and other household paper wiping / absorbing products.

BrainWave AI has identified a need for easy-to-use AI/ML for advanced analytics-driven insights and automation. It provides application-specific pre-trained models and no-code or low-code ML services that can be incorporated into current business processes. BrainWave AI specializes in LLM-based bots and software digital assistants, which can use existing organization-specific data. BrainWave AI discloses all the data used for training and does not train its model on organization-specific proprietary data.

BuckyBreakthroughs developed generic versions of silicon buckystructure applications when patents expired. The material is a form of lunar silicon resembling the structure of carbon nanotubes. It is extraordinarily strong in tension, tolerates the space environment, and can be formed into flexible strands and cables of unlimited length, or vast nets and sheets of fabric. Colors range from milky white to quartzlike transparency; properties of different forms and with various introduced impurities include thermal insulating qualities, electrical conductivity, sound transmission, adhesion, or light refraction. One variant is a bright white fabric that prevents penetration by space debris up to two inches in diameter (but offers no thermal insulation). The company was the first to develop windows made from silicon buckystructure materials for use on space structures. Windows can be ordered in any shape. With proper sealing, standard windows with no dimension greater than 3 ft. (91 cm) can retain up to 1 Earth atm pressure. If never exposed to direct sunlight, windows provide adequate radiation protection and thermal insulation.

Carbon Creations collaborates with Waste Products and Toss It To Me to receive carbon from their lunar installations, from which it makes useful products ranging from fuels to plastics. The company is also researching forms of silicon buckystructure fabric and carbon-based resins that can be used to make a rigid product similar to--but stronger and more durable than--fiberglass.

Clean Up Your Act recycles water from cleaning, kitchens, and agriculture, and delivers potable water back into space communities' safe water supply. The company also advises communities on maintenance of proper atmosphere composition, and provides air revitalization when CO_2 or other atmosphere components go out of balance. The company has an agreement with Waste Products that it can install its infrastructure simultaneously with installation of sewer lines.

Custom Cargo Accommodations produces Cargo Accommodation in Standard Space Shipping Container (CASSSC) units, compatible with standard interfaces in launch vehicles and interorbital spacecraft currently in use. CASSSCs are 30 feet (9.144 meters) long with nearly-square 15-foot (4.572 meters) cross-sections (corners of the cross-section are rounded with a 1-foot radius). Generic CASSSCs are aluminum, fully enclosed, and vented to permit pressure equalization. Special-order CASSSCs can be whatever customers choose within standard size and interfaces constraints, including pressurized, open framework, or made of composite materials.

Dirtbuilders has contracted with CalEarth (calearth.org) to be the supplier of SuperAdobe casings and robotic assemblers for extraterrestrial applications. The company makes casings for lunar and orbital applications from silicon buckystructure fabric. In lunar gravity, SuperAdobe structures can be built up to 70 feet (21.34 m) high and 50 feet (15.24 m) in diameter.

ElectroProtect builds components for circuitry that can withstand space environments, and shielding or protective boxes for components, circuits, electronics, computing equipment, and robotics that cannot be built to withstand local environments.
Extreme Survival Technologies (EST) builds spacesuits, pressurized fabric impact protection systems (e.g., airbags and restraints), and portable emergency shelters. Its most popular products are hard-shell spacesuits customized for lunar operations but frequently used for other applications; efficiencies of assembly line production enable $400,000 unit costs.

Fusion Founders serendipitously happened upon an apparently ideal combination of conditions and equipment to produce practical fusion power in 2032. The company has been busily producing power plants since that time, at its manufacturing facility in Yukon Territory. Although it can assemble large municipal power plants at customer-specified sites, its most popular product is a self-contained unit that can be shipped on a modified commercial version of a C-18 transport aircraft, and installed by local labor with supervision provided by a company engineer. The unit, weighing 200,000 pounds, includes a 17-foot diameter sphere, its 80-foot-long cooling "barn", and a support "shed". The system is shipped pre-assembled, and generates 10 MW, appropriate for non-industrial communities of about 5000 people. Fusion Founders has received several solicitations to develop a version of this unit that could be launched into space, but feels that it has achieved the theoretical limit of smallness for a fusion reactor, and cooling is not practical in space.

Hard Roll accepts ores from lunar mining operations, refines the metals, and produces rolled sheets, extruded beams, and custom shaped cast parts. The processes require approximately 10 GWh of electric power per ton of aluminum or titanium, and 2 GWh per ton for other metals.

Holey Moley adapted designs of excavation equipment used on Earth, for use in 1/6 g on the lunar surface. It has created mining equipment, trench-diggers, backhoes, dirt-movers, graders, drills, and tunnelers, and will create new machines on request. Some types of applications require more than one design solution, depending on whether local conditions enable bracing the equipment to compensate for lack of gravity. The company founder says ”rocks on the Moon are just as hard as rocks on Earth, but my machines don’t have the same weight to punch through them”.

Large Print adapted capability for 3D printing of large parts to the lunar environment. The company Hard Roll provides metal feedstock for printing parts going into spacecraft, construction equipment, and land vehicles. Large Print is experimenting with feedstocks and equipment to replicate “whole house machine” printing done with concrete on Earth.

LightWorks provides soletta and lunetta illumination for lunar and terrestrial surfaces locations. The huge orbiting structures reflect sunlight; six lunettas (one km2 each, $7M cost) in a night sky are equivalent to traditional street lighting. Larger solettas (six km2 each, $50M cost) enable solar power plants to operate all night. The devices consist of sodium-coated fabric made from lunar materials stretched over lightweight composite structures, and are placed in 2500 mile orbits.

Litigation Limiters is a law firm that created a niche market which virtually eliminates conflict-of-interest suits for its clients, who usually are companies competing for the same contracts but in need of each others' products and/or services. Litigation Limiters arranges its own contracts between providers of products and services, and customers, then serves as a go-between to supply the products and services to the customers. The company has such agreements with all of the world's diversified corporations that have significant product lines applicable to space development. Litigation Limiters charges a 2% fee on product or service value.

Lossless Airlocks has developed and sells airlocks that operate with almost no loss of atmosphere for each opening to space. Airlocks come in several sizes, including a single-person unit, a personnel transfer system that can simultaneously accommodate three people in adjacent chambers, and small versions for exposing experiments to vacuum without requiring an astronaut to go EVA. Airlocks for orbital use will be built at Bellevistat. The company also produces an interface that attaches the end of a CASSSC to an airlock, enabling unloading and loading in a pressurized environment.

Lunar Adventures provides spacesuits and guide-operated vehicles for lunar surface excursions. The vehicles provide a shirtsleeve environment for six passengers plus the guide, and enable overnight "camping". The electric-powered vehicles can be self-sufficient for up to a week in sunlight with storage for two days of power in darkness, and have a range of 200 miles per day. Passenger fees are $10,000 per person per day, or $50,000 per day for charter of a vehicle and guide. Spacesuit rentals are $8,000 per person per trip, plus $1000 per day. Company owners hope to meet with Foundation Society officials to arrange for basing their operation at Alaskol.

Magnetic Propulsion Company was founded by three professors at Princeton University who continued work with mass drivers originally started by Gerard O'Neill. The professors claim that they can develop 200,000 pounds of robots and equipment that can be operated by a crew of five and assemble a complete working two-mile long mass driver anywhere on the lunar surface in six months, using lunar materials. (A habitat and vehicles for the crew are not included in this estimate.) The company’s mass drivers provide both acceleration for launch, and deceleration for landing. The Foundation Society has contracted with the company to build a mass driver on the lunar equator. Company leaders anticipate that a mature lunar transportation infrastructure will include a few large mass drivers near major export locations, and small mass drivers for ballistic transportation between major destinations, all connected by a global monorail network.

Mirror Image makes mirrors from lunar materials that are used to reflect sunlight while allowing radiation to pass through. The standard size is 13.5 feet (4,11 m) by 30 feet (9.14 m); up to 175 panels can be shipped in a specially modified CASSSC.

Nano Solutions was established at Alexandriat to commercialize production and marketing of nanobots after techniques were developed to grow them in zero g and vacuum. The microscopic robots perform tasks at the molecular level--primarily modifying molecules to form an airtight seal on interior surfaces, fusing coatings on surfaces, and separating elements mixed in metallic asteroids Programmed nanobots sell by the ounce at roughly ten times the cost of platinum; when delivered they resemble a fine powder the customer applies as a thin layer to the working surface. Service life is one to five Earth years, depending on operating environment and application.

OrbitLink Communications was established when the Alexandriat Space Settlement was under construction, to augment traditional communications channels. It subsequently developed standard equipment adaptable to any habitable space installation. The company has made arrangements to place one of its antennas and dedicated fiber optics links on every Foundation Society settlement.

Planetary Pavers operates customized autonomous road-building equipment on Earth's Moon. The company's grading and paving machinery is solar powered, with 14 Earthdays of battery life for operations in darkness. Their capabilities include breaking rock to make gravel, and sintering regolith to create a hard and smooth road surface. After a surface is leveled, about 0.6 mile (one km) of paved surface can be completed in 24 hours. The company can provide gravel and paving materials for orbital installations.

Remotely Local Products is a spinoff of a Vulture Aviation team that commercialized manufacture of common household and office items from lunar materials. The company slogan is "give us carbon and silicon, and we'll give you home".

Scrub-A-Dub-Dub is the space-based subsidiary of a large consumer products company that started business with a new bar soap product over 200 years ago. The company was an early adopter of processes to manufacture product substitutes using in situ space resources, with an emphasis on cleaning products. The company President says "wherever people go, they need to keep themselves and their stuff and their living spaces clean".

Seals-It-All makes paints and coatings from lunar materials that provide air-tight surfaces on rock, SuperAdobe, and other solid but porous surfaces. Application is done with standard paintbrushes that can be manipulated by robots, and the surface requires two Earth days to dry before it is capable of retaining air at Earth sea level pressure.

Soil Solutions converts lunar and asteroidal dirt to non-abrasive soil for agriculture and landscaping. Products are delivered in CASSSCs, which must be unloaded in a thermally benign and pressurized environment to preserve the biota in the soil.

Stuff of Life has developed processes to make air and water from lunar materials. The company collaborates with Waste Products to recover nitrogen Air and water are shipped in sealed CASSSCs; air can be liquefied to reduce volume for shipping.

Toss It To Me recycles trash and garbage from lunar and space habitations. What it receives is diverse and variable; the company President says “what we do is almost like alchemy, and most of what we recover to sell is byproducts”. An especially valuable byproduct is methane from rotting garbage in landfills. Current processes manage to repurpose about 50% of input; a goal is to send no more than 10% to landfill. Each customer must allocate an interior area equivalent to 1% of the residential community’s land area for a building where conversion processes occur, a similar area outside the settlement for an unpressurized structure, and a third area as a landfill.

Waste Products has developed toilets and sewage handling systems that work in reduced gravity, and convert human waste to recycled water, excellent fertilizer rich in phosphorus, and a source of carbon. The systems do require that users decide before each sitting whether they will do “#1” or “#2”-- dual ops is not currently supported. The company works with developers of new lunar habitations to design and install sewer systems; it requires uninhibited access to a building site for 100 homes for one month after the site is graded and prepared for construction.

Wheels of Fortune builds vehicles for lunar exploration and long-distance cargo hauling. The vehicles are currently built on and shipped from Earth. Exploration vehicles can operate for up to three months away from civilization with a crew of six. Cargo hauling vehicles can operate autonomously (driverless) on lunar roads equipped with navigation aids, and can tow trains of up to ten trailers, each trailer supporting one filled CASSSC. Both types of vehicles are shipped in CASSSCs; an exploration vehicle completely fills a CASSSC, two tow vehicles can fit in one CASSSC, and six trailers can fit in one CASSSC.

ZAP! Industries supplies wire harnesses for electrical power distribution, and fiber optics systems for electronic signals on spacecraft. The company operates a system for zero-g manufacturing of solar cells from materials available in silicate asteroids. ZAP! sells each of these units for $40 million, not including transportation to deposit it on an appropriate asteroid, where it produces 1 x 2 foot solar panels at the rate of 1,000 per day, each of which is capable of generating 38 watts of power in Earth orbit and weighs 2 pounds, at a cost of $80 per kW (not including delivery).

ZeroFlux Innovations emerged on the market in the late 2020s, originally producing thermal control technologies for small satellite applications. The company has since pivoted to providing radiators, heat exchangers, and other thermal control solutions tailored to the needs of space settlements and large spacecraft. ZeroFlux Innovations provides comprehensive analyses of the customer’s thermal control requirements and guarantees design and assembly of the necessary equipment. Although the company’s technologies are primarily designed to be implemented in the vacuum of space, the company can size the equipment for any location if provided data on a pressurized environment’s atmospheric composition and pressure, gravitational forces, hours of sunlight exposure, and expected wind speeds and other weather conditions of the operational location.

Appendix B: Space Settlement Design Competition GLOSSARY

Many of the words and terms used in Competition materials are not part of familiar everyday usage:

air-breathing engine: propulsion plant (motor) that acquires oxidizer from the atmosphere, rather than carrying it in tanks on the vehicle (as required by rocket engines)

airlock: chamber enabling people and things to move or be moved between volumes with different pressures; like a lock in a canal, the chamber starts at the pressure that the occupant is moving from, and changes to the pressure being moved to

attitude (of a vehicle): vehicle's orientation relative to Earth, Sun, or other objects; typically used to describe a desired view, observation target, or heating environment (e.g., a "sun-facing" attitude assures that one side of the vehicle will always be hot, and the other side always cool)

avionics: literally, "aviation electronics", mostly including commanding and monitoring of systems on aircraft and spacecraft

cargo: reason a vehicle flies; stuff carried by a vehicle from its starting point (ground or on-orbit) to the vehicle's destination; can include satellites, bulk materials, construction components, or people

cargo container: standard unit in which cargo is installed; container interiors are configured for complex installations, and standard exterior container interfaces are quickly mated to the inside of a cargo vehicle (standard containers are used on ships, conventional aircraft, railroad cars, and trucks)

consumables: stuff that is used up during the course of a mission or over a period of time, and hence must be replaced; includes everything from rocket fuel to pet food to pencils

contract: legal agreement whereby a company (contractor) promises to build something or provide a service to a customer within a defined cost and schedule, and the customer agrees to pay the entire cost when the product is delivered, or partial payments over the course of a long delivery schedule

dirtside: of or referring to Earth, people living there, and things on it

down area: in a rotating space structure, the interior surfaces through which the force due to the rotation ("artificial gravity") appears to be vertical; conversely, surfaces inside a rotating space structure on which a person could stand or things could be placed, as if they were on the ground

downweight: amount of payload weight carried by a vehicle from orbit to the ground

Expendable Launch Vehicle (ELV): vehicle launched only once; typically, it sheds components (stages) during ascent, with only a small portion of the original "stack" delivered to orbit

Extravehicular Activity (EVA): excursion by spacesuited person outside of a vehicle or habitat

fabrication: manufacturing; the process of making, building, and/or assembling

fiber optics: use of glass strands to transmit light representing electronic signals; can replace copper wire with less weight and expense, and greater reliability, but cannot transmit power

fines: tiny particles of lunar, Martian, or other extraterrestrial dirt; typically called “dust”

GEO: geosynchronous Earth orbit; objects in 22,300 mile orbits rotate around the Earth at the same rate that the Earth turns on its axis; when located above the Equator, these objects appear to be stationary in Earth's sky

hypersonic flight: flight through an atmosphere at greater than five times the speed of sound (Mach 5) for that atmosphere

launch vehicle: spacecraft capable of launching or flying through an atmosphere (e.g., Earth's) in order to get into space and achieve orbit

LEO: low Earth orbit; orbital locations above Earth's atmosphere and below the Van Allen radiation belts

libration points, or L1, L2, L3, L4, L5: in orbital mechanics, when one large body (e.g., the Moon) is in orbit around another large body (e.g., Earth), there are five points in orbits around the larger body where gravitational forces balance out to enable satellites to be placed where they could not stay if the smaller of the large bodies were not present (also called Lagrangian points, for Joseph Lagrange, the mathematician who developed the theory that predicts their existence)

logistics: getting what’s needed where it’s needed when it’s needed.

low-g: acceleration environment with less than the acceleration due to gravity on Earth's surface

mass driver: device that electromagnetically accelerates small objects to very high velocities; can be utilized for efficiently launching material from airless surfaces

micro-g: accurate description of "weightlessness", the condition experienced in space when forces balance out and objects seem to "float"; true "zero-g" is theoretically not possible, because there are always tiny forces operating on all objects

on-orbit: in space, in an orbit; usually refers to an orbit around Earth

orbit: path assumed by an object in space, due to balancing or "canceling out" of accelerations due to gravity and rotation; usually the elliptical path of a small body (e.g., satellite) around a very large body (e.g., planet, moon, or star)

outweight: amount of payload weight carried by a vehicle from Earth’s vicinity outbound to another location in the solar system

overhead: the part of a budget not considered cost of work directly on a project, but charged to the customer as part of the hourly cost for direct work (i.e., a contractor is paid for each hour an engineer works on tasks directly related to the project; the customer is billed a cost for the engineer's hours that is greater than the salary paid to the engineer; the difference pays for computers, facility upkeep, janitors, utilities, and other costs needed to support the engineer's work)

payload: literally, "paying load"; cargo carried by a vehicle, for which a fee is paid in exchange for moving the cargo to its destination

payload capability: weight of payload(s) that a launch vehicle is capable of carrying to orbit

payload integration: process of safely stowing a payload (usually a satellite or complex device) on a launch vehicle and providing services (including electrical power, avionics, and thermal control) that enable the payload to survive the flight and accomplish its purpose; includes design of services, analysis of payload's ability to survive environments it will experience, and installation in the vehicle

profit: difference between the price charged by a contractor for providing a product, and the actual cost the contractor incurs to create the product

proposal: document prepared by a company or other entity, to convince a customer to select the entity to provide a certain product; it describes the company's plan for providing the product, and explains why the customer can be confident that the company has a superior design and will produce it according to the customer's requirements and within the described cost and schedule

rectenna: receiving antenna, converts directed energy (e.g., from an SPS) to electric power

Request for Proposal (RFP): document prepared by a customer, which describes features of a product they want a contractor to produce

requirements: features a customer requests to be included in the design of a desired product

returnweight: amount of payload weight carried by a vehicle to Earth's vicinity inbound from another location in the solar system

Reusable Launch Vehicle (RLV): launch vehicle that returns from its missions intact, and is designed to be maintained after flight and fly repeated missions

satellite: any object in orbit around another object; usually refers to human-made devices in orbit around large natural bodies (i.e., planets, moons, stars)

shirtsleeve: environment in a vehicle or habitat enabling humans to operate without protective gear

Single Stage to Orbit (SSTO): launch vehicle that ascends from ground to orbit without staging, or shedding of components during ascent; such vehicles contain all fuels and oxidizer they require in tanks inside their structures, and return to the ground with tanks intact (the amount of oxidizer onboard can be reduced through use of air-breathing engines during flight in the atmosphere)

solar panel: device that converts sunlight into electrical power

Solar Power Satellite (SPS): a satellite, usually very large, consisting mostly of large arrays of solar panels producing electrical power that can be converted (usually to microwave energy) and transmitted to users in other locations

solar sail: a surface, usually very large and lightweight, that makes use of pressure due to light or solar wind for propulsion

spacer: of or referring to people who live in space

spacesuit: garment providing pressure, breathing air, fluids and nutrients, waste removal, and protection from the space environment, enabling a human to move and operate on EVA

SPS: see “Solar Power Satellite”

SSTO: see "Single Stage to Orbit"

station-keeping: use of small rockets, solar sails, or other propulsion to prevent satellites from drifting out of their desired orbital locations

terraforming: process of making a planet more Earth-like, i.e., habitable by humans

upweight: amount of payload weight carried by a launch vehicle to orbit

Van Allen radiation belts: bands of radiation trapped in Earth's magnetic field, which both absorb ambient deep-space radiation and provide protection for Earth's surface, and are a hazard for satellites and humans operating within them

zero-g: see "micro-g"

Helpful resources and training files: bit.ly/InSimEdTraining2025

2024 Central Space Design Competition

Schedule Of Events

TIME	ACTIVITY	LOCATION
FRIDAY 2/7		
4:00 PM - 5:45 PM	Check in at Gilruth Center for all participants, dinner ~~available~~ Sahara Room	Ballroom
6:00 PM	Welcome, safety message	Ballroom
6:15 PM - 6:45 PM	Guest Speaker	Ballroom
6:45 PM - 7:15 PM	In Memory: Anita Gale and Norm Chaffee	Ballroom
7:15 PM - 7:45 PM	Chaperone Meeting	Gym
7:15 PM - 7:30 PM	Mars Quiz Show	Ballroom
7:30 PM - 8:15 PM	Introduction to the Competition	Ballroom
8:15 PM - 8:30 PM	Break for snacks	Ballroom
8:30 PM - 10:00 PM	Company Meetings (company assignment reveal with Tom Quayle)	Ballroom
10:00 PM - 11:00 PM	Set up sleeping areas, showers available	Ballroom and Gym
11:00 PM	LIGHTS OFF in sleeping areas	Ballroom and Gym
SATURDAY 2/8		
7:00 AM	LIGHTS ON in sleeping areas	Ballroom and Gym
7:00 AM - 7:30 AM	Pack up sleeping areas	Ballroom and Gym
7:30 AM - 8:00 AM	Breakfast	Ballroom
8:00 AM - 9:00 AM	Department Tech Sessions	Company Rooms, Ballroom
9:00 AM - 9:30 AM	Elective Training Classes	Company Rooms, Ballroom
9:30 AM - 10:00 AM	RFP Review	Ballroom
10:00 AM	Begin work on designs	Company Rooms
12:00 PM - 1:30 PM	Lunch	Ballroom and Pavilion
6:00 PM - 8:00 PM	Red Team (1 hour per company, 2 red teams)	Company Rooms
5:30 PM - 7:00 PM	Dinner	Ballroom and Pavilion
10:30 PM	Pizza	Company Rooms
11:00 PM	LIGHTS OUT in sleeping areas	Ballroom and Gym
SUNDAY 2/9		
7:00 AM	LIGHTS ON in sleeping areas	Ballroom and Gym
7:00 AM - 7:30 AM	Pack up sleeping areas, showers available	Ballroom and Gym
7:30 AM – 8:30 AM	Breakfast	Ballroom

8:30 AM	Presentation Slides Due, company room inspection	Ballroom, company rooms
9:00 AM - 9:50 AM	Company 1 presentation (35 min) + Q&A (10 min) + intermission (5 min)	Ballroom
9:50 AM - 10:40 AM	Company 2 presentation (35 min) + Q&A (10 min) + intermission (5 min)	Ballroom
10:40 AM - 11:30 AM	Company 3 presentation (35 min) + Q&A (10 min) + intermission (5 min)	Ballroom
11:30 PM - 12:20 PM	Company 4 presentation (35 min) + Q&A (10 min) + intermission (5 min)	Ballroom
12:20 PM - 2:00 PM	Lunch, judges meeting, company photos	Ballroom and pavilion
2:00 PM - 2:30 PM	Closing Ceremony and Announcement of Winning Team	Ballroom
2:30 PM - 3:00 PM	Pack buses, prepare for departure, clean up	Ballroom